Segmentation of Hepatic Tissue in Magnetic Resonance Imaging Using Fuzzy Logic

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Abstract*—*Objective: **The purpose of this paper is to describe a manually-initialized liver segmentation algorithm for in- and opposed-phase MR images and evaluate its performance.** ***Methods:*** **First, the user inputs seeds over the liver tissue, which spread in a region growing operation. Then, fuzzy rules are applied to the roughly segmented volume, in both IP/OP images. It classifies voxels into liver, inner structures and background. Finally, the results from fuzzy operations on both images are combined using morphological operations.** ***Results:*** **The proposed method was tested against 32 pairs of magnitude OP/IP MR images. The average Jaccard coefficient was 0.85, accuracy was 0.99, sensitivity was 0.93 and specificity was 0.99.** ***Conclusion:*** **The proposed method achieved results similar to the literature, while being developed in Python language as intended for use with a multi-disciplinary team. The algorithm and datasets are available in a public repository.**

***Index Terms*—Liver, segmentation, MRI, fuzzy classifier**

I. Introduction

Liver segmentation in magnetic resonance (MR) images is a challenging task of increasing interest in research. Segmented images isolate the organ from its vicinities, allowing volume calculation, 3D modeling and tracking regions of interest. These data facilitate the evaluation of several liver pathologies [1], radiotherapy and surgery planning [2].

Many segmentation schemes have been proposed for liver segmentation using computerized tomography (CT), and tests can be run and compared to publicly available databases. Magnetic resonance imaging (MRI), on the other hand, due to its several parameters, intensity inhomogeneity and artifacts does not have nearly as many algorithms or publicly available databases [3]. The liver parenchyma presents itself as a especially demanding challenge due to its intricate irrigation system and local variations in signal intensity caused by the presence of fat and iron deposits. While research advances toward precise calculation of fat fraction maps using MR, algorithms usually iterate over a region of interest (ROI) or over the whole abdomen image [4], [5]. Tissue-only segmentation avoids internal large vessels and biliary structures, which allows for more precise fat fraction calculation and opens doors for internal structures automatic segmentation.

We propose in this paper an algorithm based on fuzzy logic for segmentation of the hepatic tissue, avoiding internal major blood vessels and biliary structures and requiring only user initialization. This scheme is intended for use in MR magnitude in-phase (IP) and opposed-phase (OP) gradient-recalled echo (GRE) image pairs, a typical protocol when evaluating liver *steatosis*.

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II. Related Work

Recent works based on different imaging parameters and techniques have accomplished liver segmentation in MR images. They usually have as ultimate goal the reconstruction of a three-dimensional model of the organ for volumetry purposes, not concerning its internal structures. Chartrand et al. proposed in [3] a semi automated segmentation using Laplacian mesh optimization, while watershed and active contouring were preferred by Huynh et al [6] in an automated scheme, both using T1-weighted 3D gradient-echo images.

When segmentation is a mean to enable consistent and reliable hepatic fat fraction calculation, in-phase/opposed-phase (IP/OP) image pairs are the standard since they can provide some water-fat separation [4]. The presence of fat and iron deposits in the liver (focal *steatosis*) change the local magnetic field homogeneity and thus the signal, which appears differently in the in-phase and opposed-phase images, altering the quality of segmentation in fatty livers [7]. Nevertheless, important information can be retrieved and combined from this pair of images for an improved segmentation. Yan et al. [8] and Shen et al. [9] used IP/OP images to create atlas-based automated segmentation algorithms.

III. Method

The proposed segmentation scheme consists of three main stages applied in combination to both OP and IP magnitude gradient echo images (Fig. 1): manual initialization, fuzzy classification and post processing. During the initialization, the user inputs seeds over the liver tissue, which spread in a region growing operation. Then, fuzzy rules are applied voxel by voxel to the roughly segmented volume, in both images. This operation classifies voxels into liver, inner structures and background. In the third stage, the results from fuzzy operations over in- and opposed-phase images are combined using boolean and morphological operations and a second region growing is applied in order to assure the continuity

of the tissue. When there is discontinuity between the lobes of the liver in the image, the results are simply combined with a boolean operation.

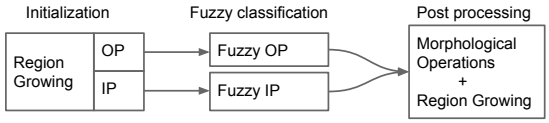


Fig. 1. Three main stages of the proposed segmentation scheme.

A. Manual Initialization

The first user inputs are the selected pair of OP-IP images and optionally a reference segmented image in order to evaluate the goodness of segmentation. A visualization window pops up showing the opposed-phase image, which usually has a better contrast and enables the user to click multiple points on the image. Some of these points will be used as seeds and they should follow a few recommendations:

- Pick seeds only within hepatic tissue, avoiding any other internal structures.
- If there are regions that belong to the liver that clearly have different intensity regions due to artifacts, focal lesions or coil gradients, include seed points in the brightest and the darkest regions too. Do not include the dark lines from chemical shift effect.
- There is no need to exaggerate the number of seeds. Only three of them will be used.

Out of all the seeds, the ones corresponding to the maximum and minimum gray intensity values are stored. Also, the average of the intensities corresponding to each seed is calculated and the seed that best approximates to it is also stored. The same points and calculations collected on the OP image are applied to the IP image, since the inhomogeneities tend to be present on both images. Three separate, simple region growing operations (4-connected, threshold 15) pre-segment the image into liver and adjacent connected regions with high, average and low signal intensity.

The outputs of this stage are: minimum, average and maximum intensity values for opposed-phased and for in-phase image seeds, and three boolean masks with `True` values where the next stage should be applied.

B. Fuzzy Classification

The fuzzy classification step receives as input a local reference value `Vm`, which can be the minimum, average or maximum value from the seeds. It then operates over the masked regions of the OP and IP images with separate set of rules due to their different nature.

In opposed-phase images with low flip angles, tissues that contain water and fat altogether are represented with less signal than tissues with water- or fat-only. Also, the signal intensity depends on the water or fat proton density. Therefore, either a healthy or steatotic liver will have less signal than its internal structure, which in general have a greater water proton density. Besides that, this sequence depicts an effect called chemical shift artifact, a dark or white band around water-containing organs surrounded by fat, such as the liver. From those information one can assume that if the intensity is much brighter or darker than the local reference, the voxel is not hepatic tissue but some other structure or chemical shift border. For in-phase images, water and fat proton densities within the same voxel add up in signal intensity when compared to water- or fat-only tissues. The rules for these images follow the same idea as for opposite-phase, except that much darker values than the local reference are considered internal structures and brighter ones are considered boundaries.

OP image rules
If pixel intensity is lower than `Vm`, then pixel is `boundary`
If pixel intensity is same as `Vm`, then pixel is `liver`
If pixel intensity is higher than `Vm`, then pixel is `vessel`

IP image rules
If pixel intensity is lower than `Vm`, then pixel is `vessel`
If pixel intensity is same as `Vm`, then pixel is `liver`
If pixel intensity is higher than `Vm`, then pixel is `boundary`

Input membership function values were defined empirically according to the intensity differences observed between tissue and non-tissue structures. The output functions for this step are crisp sets as they classify the voxels in mask images containing black (`boundary`), mid-gray (`vessel`), or white (`liver`) pixels. Please note that despite the variable is named `vessel`, it may be any internal structure other than plain hepatic tissue. The aggregation method used was `AND`, and centroid was used for defuzzification. Figure 2 depicts the membership functions for both fuzzy inputs and crisp output.

For each of the three intensities `Vm`, the fuzzy classifier outputs two images, one for OP and one for IP, with slightly different and complementary characteristics. Each has mainly three levels between 0 and 1 discerning the structures, where 0 means black and 1 means white. All six images are then combined in the post processing stage.

C. Post Processing

The algorithm's final output will only discern what is or not hepatic tissue. In order to do that, the first post processing step is to apply a logical threshold to every image result, which will convert them into logical masks, with pixel values equal to `True` when liver or `False` when not. Next, each in- and opposed-phase pair is combined using logical `AND`. Two morphological operations are applied to improve the results: binary erosion and binary opening. Erosion will enlarge the internal vessels and ducts positioned perpendicular to the slice, while opening will connect and open them when they are parallelwise. Finally, a region growing is applied using the average-intensity seed in order to "clean" the mask from any unconnected white spots resulted from misclassification.

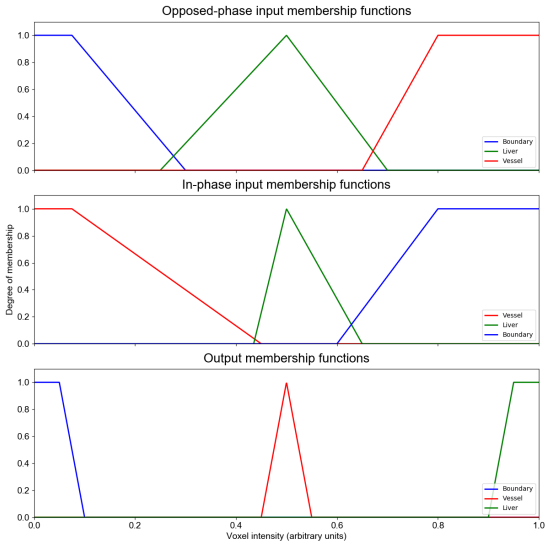


Fig. 2. Membership functions for input fuzzy OP and IP sets and output crisp set.

Finally, the resulting three masks corresponding to different `Vms` are joined using a logical `OR` operator.

D. Metrics

The algorithm was run ten times for each slice with random seeds following the aforementioned recommendations. The number of seeds varied according to the size of the liver tissue present in the evaluated slice. Jaccard similarity coefficient, accuracy, sensitivity and specificity were calculated for each run. The equations are summarized below. The average and standard deviation for these performance metrics were calculated for each dataset, and for the whole study.

Jaccard similarity coefficient

$$J(A,B) = \frac{|A \cap B|}{|A \cup B|} \tag{1}$$

Accuracy

$$ACC = \frac{(TP + TN)}{(P + N)} \tag{2}$$

Sensitivity or True Positive Rate

$$SNS = \frac{TP}{P} \tag{3}$$

Specificity or True Negative Rate

$$SPC = \frac{TN}{N} \tag{4}$$

Where A and B denote surface voxels or mask pixels, P stands for positive, N for negative, TP is true positive and TN is true negative. For Jaccard coefficient, the closer to 1 is the best.

IV. Materials

The proposed segmentation method was implemented in a Python 2.7 Jupyter Notebook [10] and executed on an laptop Intel i7-5500U @2.40Ghz computer eqquiped with 8 GB of RAM. Tippy [11], scikit-fuzzy [12], and OpenCV [13] packages were used to implement region growing, fuzzy and morphological operations. 32 pairs of magnitude OP-IP GRE liver MR image slices were sampled and manually segmented from datasets performed in a Philips Achieva MRI scanner (Philips Healthcare, Cleveland, OH, USA). The images have the following characteristics: 3 Tesla, flip angle of 30 degrees, 240x240 voxels, 26 pairs of slices acquired from a female patient with moderate *steatosis* and 6 pairs of slices containing motion artifacts, from female patient with no significant *steatosis*. The original sets are from a 2016 study approved by our institution's ethical board. Reference data were obtained by manual segmentation using ImageJ [14] and a Wacom drawing tablet (Wacom Co., Ltd., Kazo, Japan), avoiding major internal structures such as vessels and biliary ducts.

V. Results

Our method was tested on two datasets, Dataset #1 containing images of a patient diagnosed with moderate steatosis and Dataset #2 containing motion artifacts that degraded homogeneity of the image, of a patient with no significant steatosis. Before introducing and comparing the overall results with other literature approaches, we first present two image results and effects of artifacts and initialization in our method in Figures 3, 4, and 5 below.

Results for the specified metrics are presented in Table I for each set compared to their reference and for the overall study. Table II compares the Volumetric Overlap Error of a few other literature approaches and our results. The runtime changed according to the size of the liver in the specified slice, varying from 14 to 72 seconds. The choice of the the seeds also seemed to interfere in the runtime, although its effect was not measured.

TABLE I
Mean ± standard deviation of segmentation performance metrics

	Dataset #1	**Dataset #2**	**Overall**
Jaccard coeff.	0.87 ± 0.02	0.83 ± 0.04	0.85 ± 0.03
ACC	0.98 ± 0.01	0.99 ± 0.004	0.99 ± 0.01
SNS	0.93 ± 0.02	0.94 ± 0.05	0.93 ± 0.04
SPC	0.99 ± 0.005	0.997 ± 0.001	0.99 ± 0.003

TABLE II
Comparison of results from literature methods

	Jaccard coeff.	**Type†**
Yan et al. [8]	0.75	A
Shen et al. [9]	0.89	A
Proposed method	0.85	MI

† A: Automated, MI: Manual initialization

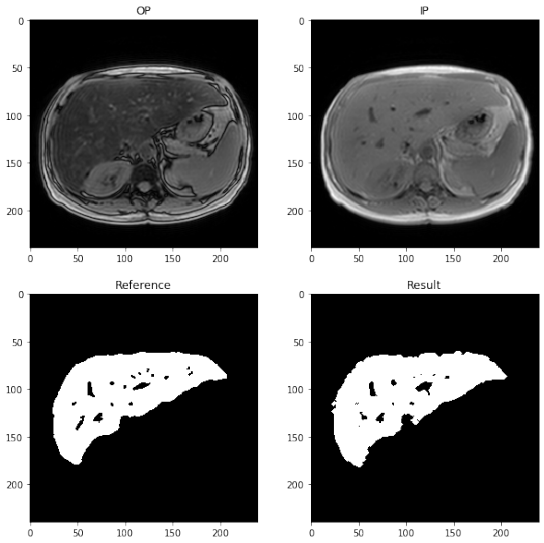


Fig. 3. Example of segmentation result with Jaccard coefficient $= 0.89$. Images from Dataset #1.

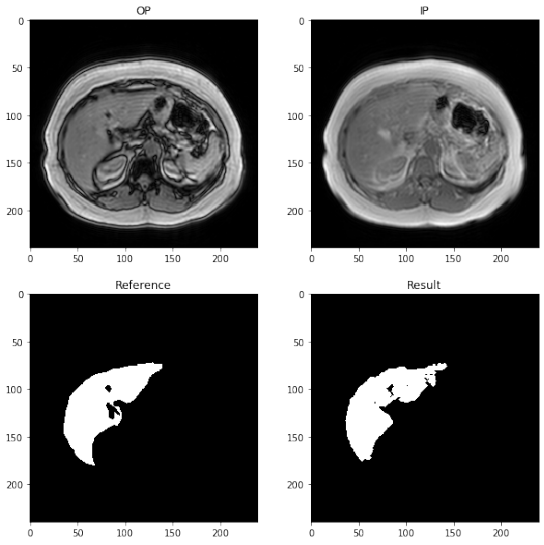


Fig. 4. Example of segmentation in image with respiratory artifacts (arched lines). Jaccard coefficient $= 0.86$. Images from Dataset #2.

VI. Discussion

From the image results, especially Figure 3 and we observe coherence in the inner structures. This is important as for fat quantification schemes it is undesired to account for vessels and biliary ducts. Also, these results can be further extended to perform three-dimensional segmentation that includes main vessel paths. Figure 4 has respiratory artifacts causing changes in signal intensity, which was overcome by the multiple-seed initialization. The choice of the seeds was unfortunate in Fig 5 though, causing under segmentation once the algorithm did not identify the increase in signal caused by the proximity of the abdomen coil as part of the liver. The `AND` operation used to join OP and IP image results also means that well

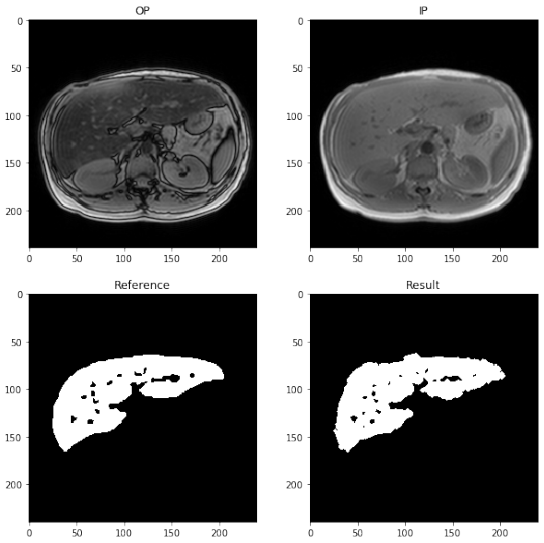


Fig. 5. Example of under segmentation (top left) caused by inhomogeneity in signal, due to coil proximity. Internal structures are preserved. Jaccard coefficient $= 0.86$. Images from Dataset #1.

segmented boundaries in will lose area when joined. That is a compromise we decided to take instead of an `OR` operation that would preserve borders but wipe internal structures.

Out of the performance metrics in Table I, Jaccard coefficient is higher in the motion-free image, as expected. Also, the presence or absence of fat did not result as an issue. Low standard deviation values may indicate agreement between the 10-run for each slice and between slices, despite no formal agreement or reliability statistics were performed. High values for specificity (0.99 ± 0.003) may be associated to the large easily identifiable background area, while sensitivity values are lower especially in the slices where the liver has little area to be be accounted for, increasing the weight of each voxel. False negatives also can occur as the final internal vessels most of the time appear smaller than the reference. Accuracy follows the trend of sensitivity while has its value increased by true negatives. A reason that makes the segmentation of vessels and small internal structures difficult is that OP and IP are not always perfectly registered, in such way that the first `AND` operation will reduce its size, hence the need for other morphological operations. Comparing our scheme performance to other authors in Table II, our Jaccard coeffient does not fall much behind Shen et al. [9] and is higher than Yan et al. [8]. The first is a multi-organ segmenter and the latter is intended for 3D usage, both looking at fat fraction calculation.

Our algorithm was implemented in Python language in order to be easily accessible to non-programmers and to test a novel, simple methodology that would explore the characteristics of OP/IP images. Hence, runtime was not a concern at this development stage. The authors recognize that the processing time can be considered too long for clinical routine and believe that if implemented in a lower-level language it can be improved. We are committed to present here a reproducible

study, providing our code easily accessible in a repository [15] along with the datasets after paper publication.

VII. Conclusion

We introduced an algorithm that relies on simple ideas intrinsic to the properties of in- and opposed-phase MR images. Its metrics do not fall behind when compared to other algorithms presented in literature despite being in its initial phase. Its use is intended to facilitate fat fraction calculation, allowing volume rendering considering the hepatic inner structures in future. We are providing the algorithm under GPL v3.0 as well as the datasets in a public repository for ease of reproduction and contribution.

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